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Solid-State Consolidation of EN AW-6005A Machining Chips: Feasibility and Influence of Rolling Parameters

Master's Thesis in
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Introduction

Aluminium occupies a central position in contemporary engineering practice because it combines low density with a favourable strength-to-weight ratio, good formability and corrosion resistance, and broad alloy and temper design space.
These attributes underpin its extensive use in transportation, building envelopes, packaging, and electrical applications, and they make aluminium a strategic material in the transition toward lighter and more energy-efficient products [3].
At the same time, the environmental profile of aluminium is strongly conditioned by the upstream production route. Primary aluminium derived from bauxite via the Bayer process and Hall-Héroult electrolysis is highly energy intensive and contributes materially to global greenhouse-gas emissions, accounting for roughly 1 % of global anthropogenic emissions and several percent of industrial electricity consumption [3, 17].
In this respect, the sustainability challenge is not whether aluminium can be recycled, but how to increase recycling efficiency and circularity while preserving alloy quality and limiting energy demand and emissions [4, 17].

Conventional aluminium recycling is largely based on remelting.
Although secondary production is widely recognised as significantly less energy demanding than primary production (requiring only a small fraction of the energy input and associated emissions), melting-based routes remain intrinsically affected by yield losses and quality constraints [4].
For fine and light-gauge scrap, such as machining chips, the high surface-to-volume ratio promotes rapid oxidation and increases the fraction of metal that reports to dross, with oxidation losses that can reach the order of tens of percent if not properly controlled [1, 11, 27].
In addition to these direct material losses, remelting requires repeated heating cycles, fluxing and melt treatment steps, and it is sensitive to contamination and compositional drift, especially when scrap streams are mixed or poorly sorted.
From an industrial perspective, this translates into economic penalties (lower recovered metal yield, higher processing cost) and environmental burdens in the form of unnecessary energy input and associated emissions, which become particularly pronounced for dispersed

manufacturing scrap streams [1, 10].

These limitations have motivated the development of Solid-State Recycling (SSR) routes for aluminium, in which chips or other solid scrap forms are converted into fully dense products without passing through the liquid phase.
SSR relies on thermomechanical consolidation mechanisms (pressure, temperature, plastic deformation and contact time) to disrupt surface films and create intimate metal-to-metal contact [1, 27].
A key scientific issue is the presence of the native aluminium oxide layer, which is chemically stable and forms rapidly in air; therefore, successful consolidation requires sufficiently large plastic and/or shear strain to fracture and disperse oxides, expose fresh metallic surfaces, and enable metallurgical bonding across former chip boundaries [12, 25].
Within this general framework, a broad family of processes has been explored for aluminium chips, including extrusion-based routes (direct hot extrusion and shear-intensified variants), severe plastic deformation approaches, and friction-stir-based consolidation concepts, among others [1, 12].
The unifying objective is to convert a discontinuous, oxide-covered particulate feedstock into a continuous bulk or semi-finished form (billets, rods, wires or sheets) while minimising material losses and reducing the thermal budget relative to remelting [8, 23].

From a sustainability standpoint, the relevance of SSR is grounded in the large gap between the energy intensity of primary aluminium and the energy requirement of recycling routes. Primary aluminium production is highly energy-intensive, requiring approximately 45 kWh/kg and generating about 12 kg CO_2/kg of aluminium.
In contrast, conventional secondary production through remelting reduces the energy demand to roughly 2.8 kWh/kg and emissions to about 0.6 kg CO_2/kg [4]. Although this represents a drastic improvement compared to primary production, remelting-based recycling of machining chips remains affected by significant drawbacks [7].

Due to high surface-to-volume ratio of chips, oxidation losses during melting can be substantial, with overall metal recovery in conventional recycling of turnings reported to be as low as $\sim 54\,\%$ [23]. Furthermore, remelting requires chip pre-treatment (cleaning, drying, de-oiling), fluxing operations, and high-temperature furnaces, all contributing to additional energy consumption, process complexity, and environmental impact [13].

SSR aims to overcome the intrinsic limitations of melting-based recycling of machining chips by avoiding the liquid phase and, consequently, reducing oxidation-driven metal losses, dross formation and fluxing requirements. This is particularly relevant for aluminium chips, whose high surface-to-volume ratio makes them highly susceptible to oxidation during

remelting. By relying instead on pressure, temperature and plastic deformation, solid-state routes can simplify the recycling chain and reduce the thermal burden associated with conventional furnace-based processing.
Recent comparative life-cycle assessments indicate that chip-based solid-state recycling can achieve a lower environmental footprint than remelting, provided that the process is implemented with careful control of equipment efficiency, processing time and overall energy demand [4, 12, 23].

Despite its promise, SSR of aluminium is not a purely "processing shortcut"; it is a material engineering problem in which process conditions determine interfacial integrity, microstructure evolution and property uniformity.
Consolidated products may exhibit heterogeneity associated with temperature and strain gradients, non-uniform oxide fragmentation, and local variations in bonding quality, which can manifest as spatially varying hardness, strength and ductility [1, 12, 16].
Consequently, any SSR route must be assessed not only in terms of densification and yield, but also regarding microstructural homogeneity, defect populations, and the reproducibility of mechanical performance.
In parallel, industrial scalability requires that consolidation routes be compatible with realistic scrap handling (chip morphology, cleanliness, lubricant residues), and that they deliver semi-finished geometries suitable for downstream forming and heat treatment without eroding the energy and yield advantages that motivate SSR in the first place [1, 10].

Within this context, the present work investigates the solid-state recycling of aluminium machining chips through a specific compaction and hot-rolling route.
The study is structured around three main objectives. The first is to assess the technical feasibility of consolidating aluminium chips into dense sheet material by means of thermo-mechanical processing; the second is to evaluate the influence of key process parameters and post-processing conditions on mechanical properties and material homogeneity; the third is to identify the operational window and technological limits of the proposed process, with particular attention to processing temperature and achievable consolidation.

1 | State of the Art

1.1 Aluminium

Over the past two decades, aluminium production has expanded rapidly in response to the growing demand for lightweight, high-performance materials in transportation, construction, packaging, and energy systems.
Global primary aluminium output has increased almost threefold since the early 2000s, with China accounting for more than half of total production, making aluminium the most produced non-ferrous metal worldwide [3].
This growth has reinforced aluminium's strategic role in modern manufacturing, but it has also intensified concerns related to energy consumption, greenhouse gas emissions, and material efficiency across the entire value chain [4, 17].

From a sustainability perspective, aluminium presents a paradox.
On the one hand, its low density, corrosion resistance, and recyclability enable significant life-cycle benefits, particularly in transport applications where weight reduction translates directly into lower fuel consumption and emissions.
On the other hand, primary aluminium production remains highly energy intensive. The Bayer process followed by Hall-Héroult electrolysis require large quantities of electricity, resulting in a substantial carbon footprint that places the aluminium industry among the major industrial contributors to global CO_2 emissions [7].

The environmental burden of primary aluminium is therefore strongly connected to the energy source used for electrolysis. This aspect is particularly relevant when comparing primary production, conventional secondary production, and solid-state recycling routes, because the apparent advantage of a recycling technology can change depending on the electricity mix and on the number of thermal steps required [3, 4, 17].

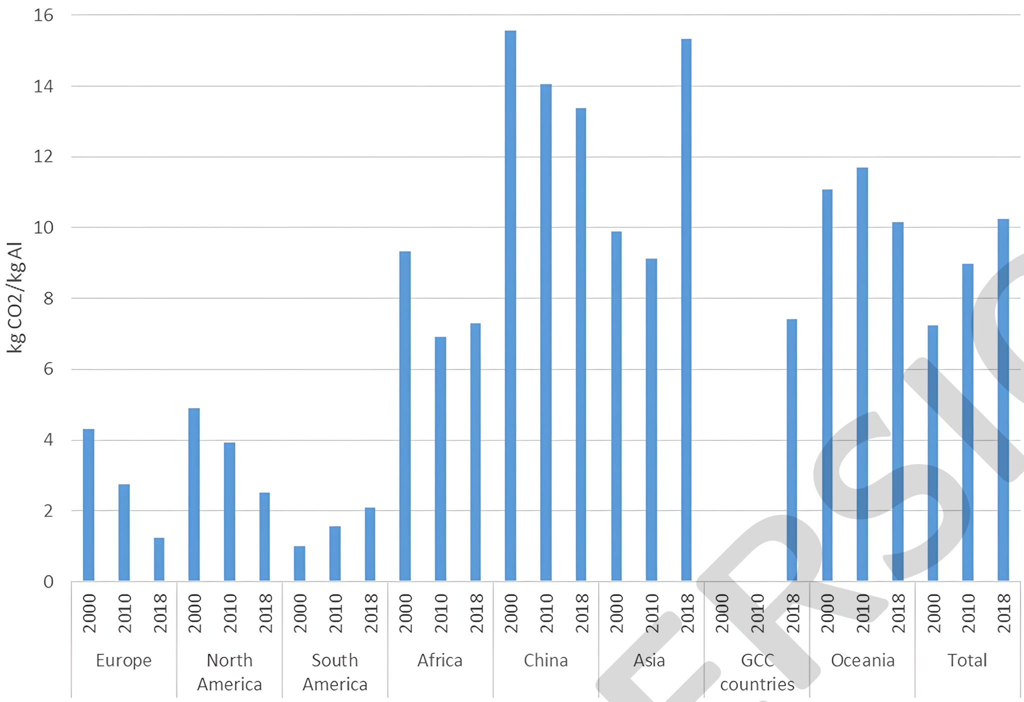

Figure 1.1: Regional variation in the carbon footprint of power production for aluminium electrolysis [22].

Therefore, recycling has become a central pillar of aluminium sustainability strategies.
Conventional aluminium recycling is predominantly based on remelting of scrap.
Secondary production drastically reduces energy demand compared to primary routes and is therefore widely regarded as an environmentally favourable option [4, 17].
However, remelting-based recycling exhibits intrinsic limitations when applied to fine and fragmented scrap, such as machining chips and turnings.
Due to their high surface-to-volume ratio, aluminium chips oxidise rapidly during handling and melting, leading to significant metal losses in the form of dross.
In addition, contamination from cutting fluids and alloy mixing complicates scrap management and often necessitates dilution with primary metal to maintain composition control [1].
Studies report that, in the case of aluminium turnings, cumulative losses along melting, casting, and subsequent processing stages may reduce the effective metal recovery to little more than half of the original scrap mass [23].

These drawbacks have driven increasing research interest toward alternative recycling routes capable of preserving material value while reducing energy input and metal losses.
Within this context, SSR of aluminium has emerged as a promising approach.
SSR encompasses a family of processing routes in which aluminium scrap, typically in the

form of chips or granulated material, is directly consolidated into bulk or semi-finished products without passing through the liquid phase.
By avoiding melting, SSR aims to suppress oxidation losses, limit emissions associated with thermal processing, and shorten the recycling chain [8, 27].

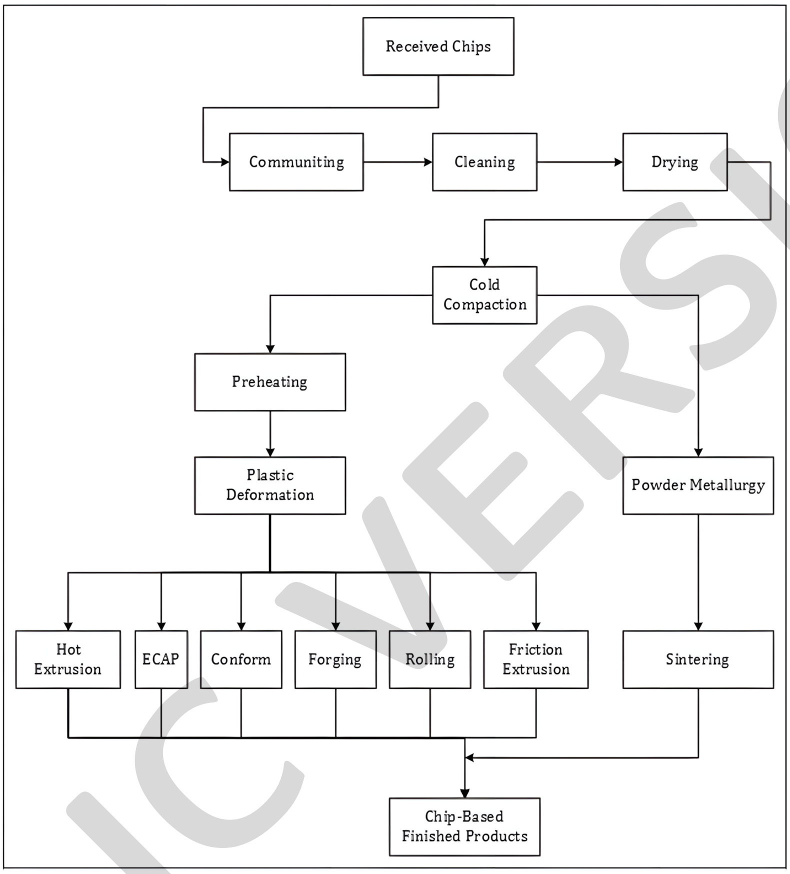

Figure 1.2: Main families of solid-state recycling methods for light-metal chips [23].

The common feature of these routes is the direct conversion of a discontinuous chip feedstock into a coherent product through pressure, temperature and plastic deformation. The individual techniques differ mainly in the way strain is introduced: extrusion provides high hydrostatic pressure and large shear, severe plastic deformation methods impose very high accumulated strain, while rolling-based routes aim to exploit existing sheet-forming infrastructure [8, 23, 24].

The fundamental challenge of solid-state recycling of aluminium lies in the presence of the native oxide layer that forms instantaneously on aluminium surfaces. This oxide is thermodynamically stable and mechanically strong, acting as a barrier to metallurgical bonding between individual chips. Consequently, successful SSR requires the application of sufficient pressure, temperature, and plastic deformation to fracture and disperse oxide films, expose fresh metallic surfaces, and promote solid-state welding across chip interfaces.

The quality of inter-chip bonding has been consistently identified as the primary factor governing the density, mechanical properties, and structural integrity of solid-state recycled aluminium products [25, 27].

A wide range of solid-state consolidation techniques has been investigated for aluminium chips. Among these, hot extrusion remains the most extensively studied and industrially mature process.
In this route, pre-compacted chip billets are subjected to large plastic strain under elevated temperature, generating high hydrostatic pressure and intense shear that favour oxide disruption and bonding.
Numerous studies have shown that hot-extruded aluminium chips can achieve relative densities exceeding 95–98 % and mechanical properties comparable to, and in some cases exceeding, those of conventionally extruded wrought alloys [23].
The effectiveness of extrusion-based recycling is strongly influenced by extrusion ratio, die geometry, processing temperature, and pre-compaction strategy, all of which control the strain path and bonding conditions within the material [23].

Beyond conventional extrusion, Severe Plastic Deformation (SPD) techniques such as Equal Channel Angular Pressing (ECAP), cyclic extrusion-compression, high-pressure torsion, and friction-based processes have been explored as SSR routes for aluminium.
These methods impose extremely high shear strains, leading to pronounced grain refinement and, in favourable cases, near-complete elimination of chip boundaries.
While SPD-based approaches have demonstrated excellent microstructural refinement and bonding at laboratory scale, their industrial applicability remains limited by process discontinuity, tooling complexity, and restricted throughput [21, 23, 24].

More recently, friction-based consolidation routes, including friction stir extrusion and related processes, have gained attention for aluminium chip recycling. These techniques combine frictional heating with intense plastic flow, enabling continuous consolidation of chips into rods or wires with fine, equiaxed misstructures.
Reported results indicate that, under optimized conditions, friction-based SSR can produce defect-free aluminium products with good mechanical performance, while operating at temperatures significantly lower than melting. Nevertheless, sensitivity to process parameters and challenges in scale-up continue to limit widespread industrial adoption [12, 27].

In parallel with process development, increasing attention has been devoted to the role of chip morphology, surface condition, and contamination in solid-state recycling. Machining parameters strongly influence chip size, shape, and microstructure, which in turn affect compressibility and bonding behaviour during consolidation.

Chips generated under severe deformation may already exhibit refined microstructures and high dislocation densities, potentially facilitating bonding during SSR [27].
Conversely, residual lubricants and mixed alloy streams introduce variability that complicates process control. Recent reviews emphasise that minimising chip cleaning and preparation steps is essential to preserve the environmental advantages of solid-state routes, yet this must be balanced against the need for reliable and reproducible material properties [10, 20].

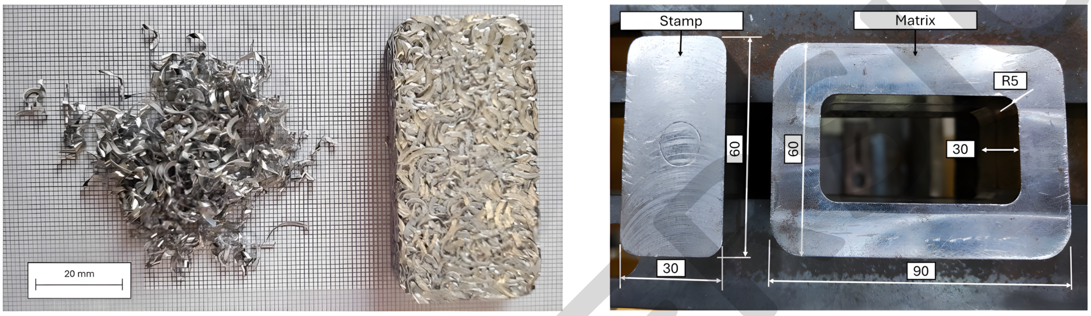

Figure 1.3: Loose AA6063 chips, compacted billet, and die geometry used for cold compaction [18].

Chip morphology is especially important for compaction-based recycling routes. Curled or irregular chips may mechanically interlock and create a handleable green compact, but they can also trap voids and generate local density gradients. For this reason, cold compaction is not only a preliminary handling step; it partly defines the initial contact area, porosity distribution and deformation path that the subsequent hot rolling stage must overcome [10, 20, 27].

From a life-cycle perspective, SSR of aluminium has been identified as a route with significant potential to reduce both energy consumption and CO_2 emissions compared to conventional remelting, particularly for machining scrap.
Life-cycle assessments indicate that the avoidance of melting and dross formation can yield substantial environmental benefits, provided that consolidation processes are energy-efficient and integrated within existing manufacturing chains [12].

However, the actual sustainability performance of SSR depends strongly on process scale, equipment efficiency, and the degree to which solid-state recycled products can directly substitute conventionally produced semi-finished aluminium forms [4, 17].

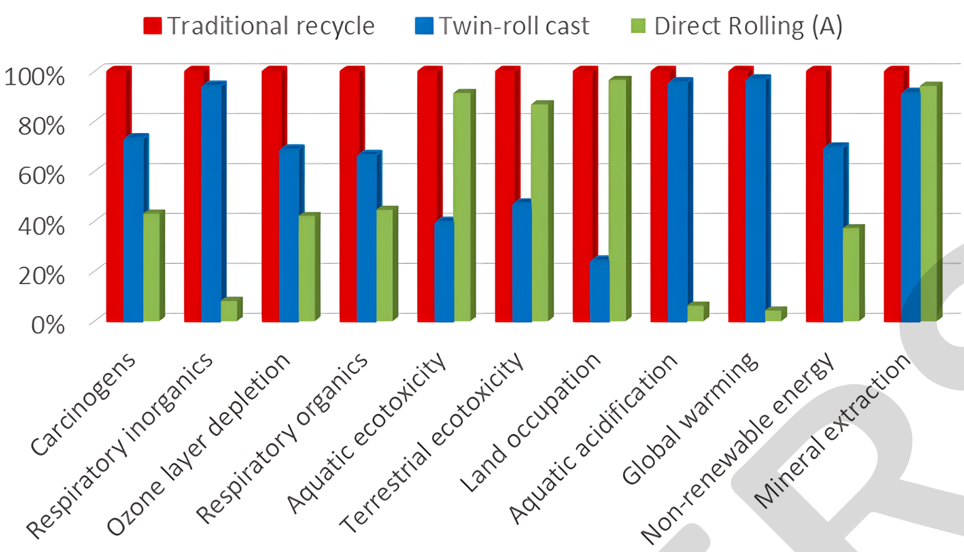

Figure 1.4: Impact-category comparison between traditional recycling, twin-roll casting and direct rolling solid-state recycling [4].

For direct rolling routes, life-cycle results must be interpreted together with process yield and product quality. Avoiding remelting can reduce dross-related losses and eliminate casting, but hot rolling, reheating, tooling and trimming still contribute to the total impact.
Consequently, the most attractive solid-state route is not simply the one with the highest deformation, but the one that reaches adequate consolidation and mechanical performance with a limited and reproducible process chain [4, 12].

1.1.1 EN AW-6005A Aluminium Alloy

Among the wrought aluminium alloys, the 6000 series (Al-Mg-Si system) represents one of the most widely used families for structural applications. In these alloys, magnesium and silicon are the principal alloying elements and combine to form the intermetallic phase Mg_2Si.
When subjected to appropriate solution treatment and ageing, this phase precipitates in a controlled manner, enabling precipitation hardening and providing the characteristic strength of the alloy family.

6000-series alloys exhibit a favourable balance between mechanical strength, ductility and corrosion resistance, together with good weldability and excellent extrudability.
Compared to high-strength 2000 and 7000 series alloys, they offer lower peak strength but improved formability and better resistance to environmental degradation, which makes them particularly suitable for transportation, building, and marine applications [15].

1.2 Solid-State Diffusion Bonding

Solid-State Diffusion Bonding (DB) is a joining process in which two metallic surfaces are bonded at temperatures typically between 0.5 and 0.8 T_m (absolute melting temperature) under an applied pressure and for a controlled holding time, without macroscopic melting of the base material [2].
Bond formation occurs when the mating surfaces are brought into atomic contact, enabling interdiffusion and the establishment of metallic bonds across the interface [25, 28].
According to the definition adopted by the International Institute of Welding, diffusion bonding produces a monolithic joint through local plastic deformation at elevated temperature, which promotes interdiffusion at the contacting surface layers [14].

From a metallurgical standpoint, diffusion bonding of aluminium is fundamentally more challenging than that of many other engineering metals. While steels, copper, titanium, and zirconium alloys possess oxides that either decompose or dissolve into the bulk at bonding temperatures, aluminium is covered by a continuous, chemically stable, and insoluble Al_2O_3 layer that persists throughout the entire solid-state temperature range [25]. This oxide film constitutes the primary barrier to achieving reliable and high-strength diffusion bonds in aluminium.

The diffusion bonding process is commonly described as occurring in three sequential but overlapping stages.
In the initial stage, applied pressure and elevated temperature induce local plastic deformation of surface asperities, increasing the real contact area between the faying surfaces. This stage is dominated by micro-plastic flow and asperity flattening rather than bulk deformation [28].
In second stage, thermally activated diffusion and creep mechanisms contribute to the shrinkage and elimination of interfacial voids, progressively increasing bonded area.
In the final stage, long-range atomic diffusion across the interface leads to grain boundary migration and, in the case of similar materials, grain growth across the bond line, resulting in a joint that may approach the microstructural continuity of the parent material [28].

For aluminium alloys, the effectiveness of these stages is strongly limited by the presence of the oxide layer. Even when nominally flat and polished surfaces are used, real contact initially occurs only at discrete asperities, and the oxide film prevents direct metal-to-metal contact unless it is locally disrupted or fragmented [25].

One of the earliest and most extensively studied approaches to overcome the aluminium oxide barrier is the imposition of significant plastic deformation during bonding. The

aluminium oxide film exhibits extremely low ductility (typically only a few percent) compared to the underlying metal.
Consequently, when sufficient plastic strain is imposed on the parent alloy, the oxide layer fractures into discrete islands, exposing fresh aluminium and enabling metallic bonding at localised contact points [25].
Experimental studies consistently reported that a minimum plastic deformation on the order of $\sim 40\,\%$ is required to obtain bonds with reasonable and reproducible strength in aluminium alloys [14, 25].

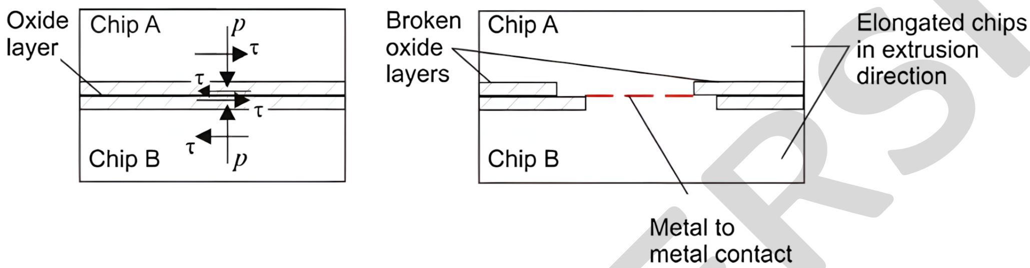

Figure 1.5: Schematic representation of oxide-layer rupture and metal-to-metal contact during deformation [27].

This mechanism is directly transferable to aluminium chip consolidation.
Each chip is surrounded by an oxide film, and bonding can occur only where deformation breaks this film and creates local metallic contact. The role of temperature is to lower the flow stress and assist diffusion and recovery, but temperature alone cannot replace the need for oxide fragmentation by plastic strain [25, 28].

Because such high deformation levels are often incompatible with component dimensional tolerances or functional requirements, alternative strategies have been investigated. One of these is the deliberate increase of surface roughness prior to bonding. Rougher surfaces concentrate stress at asperity tips under a given applied pressure, thereby promoting localised micro-plastic deformation and more effective rupture of the oxide film.
For aluminium alloys, several studies have shown higher bond strengths for roughened surfaces compared to mirror-polished ones, precisely because of enhanced oxide fragmentation at asperity contacts [25].

Given aluminium's high affinity for oxygen, diffusion bonding is commonly performed under high vacuum or inert gas atmospheres to prevent re-oxidation of freshly exposed metal surfaces after oxide rupture. If oxygen is present during or after oxide fragmentation, rapid re-formation of Al_2O_3 can occur, negating the benefits of plastic deformation and severely degrading bond quality [25].
Advanced surface preparation techniques, including in-situ ion beam cleaning and ultra-

high-vacuum deposition of interlayers, have demonstrated near-parent-metal bond strengths, but their complexity and cost restrict their applicability to laboratory-scale or highly specialised applications [25].

Although, strictly speaking, Transient Liquid Phase (TLP) diffusion bonding is not a purely solid-state process, it is often discussed alongside solid-state diffusion bonding due to the similarity of the final bond formation mechanism. In TLP bonding, a thin interlayer with a lower melting point or eutectic-forming capability is introduced between the faying surfaces.
Upon heating, a thin liquid layer forms at the interface, significantly improving surface contact and assisting in the disruption or engulfment of oxide films. Continued interdiffusion at constant temperature leads to isothermal solidification of the liquid, distinguishing TLP bonding from brazing, where solidification occurs during cooling [25, 28].
For aluminium alloys, TLP bonding has demonstrated higher and more reproducible joint strengths than conventional solid-state diffusion bonding, precisely because the presence of a liquid phase accelerates oxide fragmentation and redistribution. Nevertheless, oxide particles and impurities are typically pushed toward the bond centreline during isothermal solidification, resulting in planar interfaces that still limit joint strength and fatigue performance compared to the base material [25].

Despite decades of research, conventional solid-state diffusion bonding of aluminium alloys has not achieved the same level of industrial reliability as it has for metals with less stable oxides. Reported bond strengths often exhibit significant scatter, and fatigue performance is generally inferior to that of the parent material, which as limited widespread industrial adoption [25].
Non-conventional approaches (such as bonding near the solidus temperature, or applying complex thermo-mechanical cycles) can improve bond integrity but at the expense of narrow process windows, complex equipment, and limited component geometries [25].

1.3 Roll Bonding

Roll bonding (RB) is a solid-state joining and manufacturing process in which two or more aluminium sheets, plates, foils, or compacted layers are passed between counter-rotating rolls under sufficient normal pressure to induce bonding through severe plastic deformation. Bond formation occurs without bulk melting and relies on a combination of oxide disruption, intimate metal-to-metal contact, and short-range atomic diffusion at the interface.
Within the broader family of solid-state diffusion bonding processes, roll bonding is generally regarded as a deformation-assisted variant, where the rolling operation itself

provides both the pressure and the plastic strain required for bonding [28].

Among solid-state joining techniques applied to aluminium, RB has attracted sustained industrial and academic interest due to its relative simplicity, continuous operation, high productivity, and compatibility with conventional rolling infrastructure.
In contrast to batch diffusion bonding, RB is inherently scalable and well suited for sheet- and laminate-based products. Industrially, cold roll bonded aluminium clads are already widely used for corrosion protection, thermal management, and functional surface layers, demonstrating the technological maturity of the process for aluminium systems [9].

For a metallurgical perspective, the fundamental challenge in aluminium roll bonding remains the presence of the stable Al_2O_3 surface film. As in diffusion bonding, this oxide must be disrupted to allow metallic bonding.
In RB, oxide fracture is achieved primarily by intense plastic deformation at surface asperities during rolling, rather than by long diffusion times.
Consequently, bonding is strongly dependent on deformation-controlled parameters rather than purely thermal ones, especially in cold and warm roll bonding regimes [25, 28].

Roll bonding of aluminium is conventionally classified according to the processing temperature. Cold Roll Bonding (CRB) is carried out below the recrystallization temperature of aluminium. After stacking and surface preparation, the sheets are rolled until a critical thickness reduction is achieved. Although diffusion is limited at low temperature, the severe deformation generates significant local heating and high interfacial stresses, which promote asperity contact and oxide fragmentation.
CRB is technologically attractive because it avoids furnace heating and associated thermal effects, but it requires very careful surface preparation, high rolling reductions, and controlled environments, since adhesion-dominated bonding mechanisms are highly sensitive to surface contamination and oxide integrity [25].

Hot Roll Bonding (HRB) is performed at temperatures above the recrystallization temperature of aluminium, often following a preheating step that may be conducted in air, inert atmosphere, or vacuum depending on oxide control requirements. In this regime, plastic flow is easier, rolling forces are reduced, and short-range diffusion and recovery processes assist bond formation.
A commonly used intermediate condition is warm roll bonding, where temperatures approach but do not significantly exceed recrystallization. Compared to CRB, HRB offers a wider processing windows and more robust bonding, at the expense of additional thermal input and potential microstructural evolution such as recovery or recrystallization [28].

Accumulative Roll Bonding (ARB) represents a special case in which roll bonding is

combined with repeated stacking, surface treatment, and rolling cycles to impose extreme plastic deformation. While ARB is primarily exploited to produce ultrafine-grained aluminium sheets, a metallurgical bond is formed at each cycle between stacked layers, and therefore the process is intrinsically a roll bonding technique. Heat treatments may be applied between cycles to tune microstructure and work hardening.
Although most ARB studies focus on grain refinement rather than joint integrity, the formation of sound interfacial bonds confirms the applicability of roll bonding principles under severe strain conditions.

Across all roll bonding variants, literature consistently identifies thickness reduction as the dominant process parameter. A minimum critical reduction, often referred to as a threshold reduction R_t, must be exceeded for bonding to occur. Below this value, deformation is insufficient to fracture the oxide layer and close interfacial gaps.
Once R_t is reached, bond strength increases sharply with further reduction due to enhanced oxide disruption and increased bonded area. This behaviour has been reported consistently for aluminium systems and mirrors the minimum deformation requirement observed in solid-state diffusion bonding [25].

Rolling speed plays a secondary but non-negligible role. Higher rolling speeds increase productivity and can raise the interfacial temperature through deformation heating, potentially improving bonding. However, increased speed reduces the effective contact time available for diffusion and stress relaxation at the interface [26].
As a result, rolling speed influences both the magnitude of R_t and the resulting interfacial microstructure, and must be optimized as a compromise between thermal assistance and bonding time.

Surface preparation is particularly critical for aluminium roll bonding. Mechanical roughening, commonly achieved by scratch brushing, increases surface roughness and promotes localised micro-plastic deformation at asperity contacts.
Experimental comparisons between metals with and without stable oxides have demonstrated that, for aluminium, surface roughness is more effective in enhancing bond strength than relying on macroscopic deformation alone. This observation is fully consistent with diffusion bonding studies, where oxide fragmentation at asperities governs bond initiation [25].

Heat treatments, both prior and after rolling, also influence bond quality. Pre-rolling annealing can soften aluminium alloys, reduce work hardening from prior processing, and facilitate oxide rupture during rolling.
Post-rolling heat treatments may improve bond strength by relieving residual stresses,

reducing hardness gradients, and enhancing short-range diffusion across partially bonded interfaces. However, excessive thermal exposure can promote grain growth or degrade laminate integrity and therefore heat treatments must be carefully tailored to the alloy and process route [28].

Typical defects in aluminium roll bonding are primarily interfacial in nature. Interfacial gaps and unbonded regions arise from insufficient plastic deformation, inadequate surface preparation, or failure to exceed the critical reduction R_t. Oxide remnants trapped at the interface are a recurrent feature, particularly in CRB.
In HRB, additional defects such as porosity and interfacial cracking may develop due to thermal mismatch during cooling or vacancy fluxes associated with differential diffusion rates, leading to Kirkendall-type voids. Although intermetallic compound formation is a major concern in dissimilar-metal roll bonding, for aluminium-aluminium systems this issue is largely absent, and defect control is dominated by deformation and oxide management [9].

In recent years, roll bonding has also been explored as a direct SSR route for aluminium machining chips. The underlying motivation is to combine the high productivity of rolling with the energy and emission savings of solid-state recycling.
Studies on hot rolling of compacted aluminium chips have demonstrated that, following appropriate chip annealing and compaction, roll bonding can produce layered laminates with broken oxide networks at chip-chip interfaces, indicative of effective diffusion bonding [16, 19].
Microstructural observations consistently show fragmented oxide films aligned along rolling direction, confirming the dominant role of deformation-assisted bonding mechanisms.
Mechanical testing of such laminates has shown yield and tensile strengths meeting or exceeding standard requirements for wrought aluminium products, although fracture surfaces often reveal mixed ductile-brittle features associated with incomplete consolidation [5, 6].

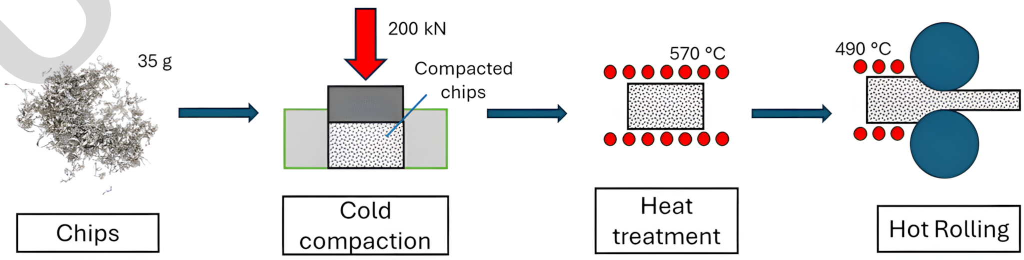

Figure 1.6: Schematic direct-hot-rolling route: chips, cold compaction, heat treatment and hot rolling [18].

Additional process developments, such as the use of sacrificial or protective envelopes around compacted chips assemblies, have been proposed to enable lubrication during rolling, reduce roll wear and improve surface finish without compromising interfacial bonding. When the envelope alloy is compatible with the base aluminium, it becomes part of the final laminate and contributes to the overall mechanical response [6].

Direct hot rolling of compacted chips can be considered a particular extension of roll bonding. Instead of joining two prepared sheets, the process must bond a large number of irregular chip interfaces while simultaneously closing pores and transforming the compact into a sheet.
This makes total reduction, reduction per pass, billet temperature and friction at the roll-material interface particularly important, because they control both oxide disruption and material flow [5, 16, 18].

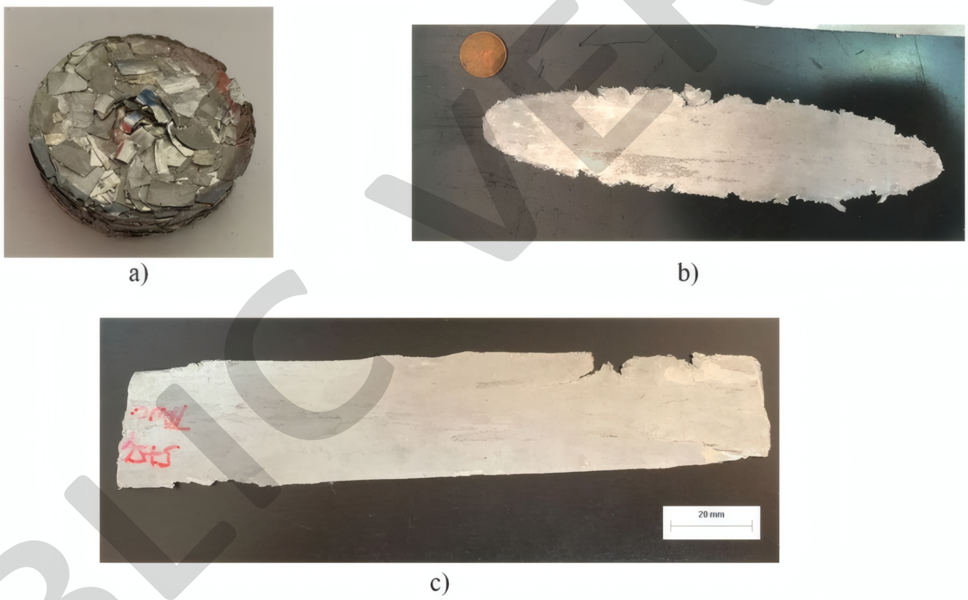

Figure 1.7: AA5754 compacted chips before recycling, sample after hot rolling and sheet after cold rolling [18].

Additional process developments, such as the use of sacrificial or protective envelopes around compacted chips assemblies, have been proposed to enable lubrication during rolling, reduce roll wear and improve surface finish without compromising interfacial bonding. When the envelope alloy is compatible with the base aluminium, it becomes part of the final laminate and contributes to the overall mechanical response [6].

Microstructural observations of direct-hot-rolled chip materials commonly reveal elongated prior chip boundaries and oxide-rich layers aligned with the rolling direction. When these layers are fragmented and discontinuous, they indicate that deformation has promoted

local matrix continuity; when they remain continuous, they may act as weak interfaces and preferential crack paths.
Therefore, density measurements, optical microscopy, hardness profiles and tensile tests must be considered together when assessing the quality of a solid-state recycled sheet [1, 12, 16].

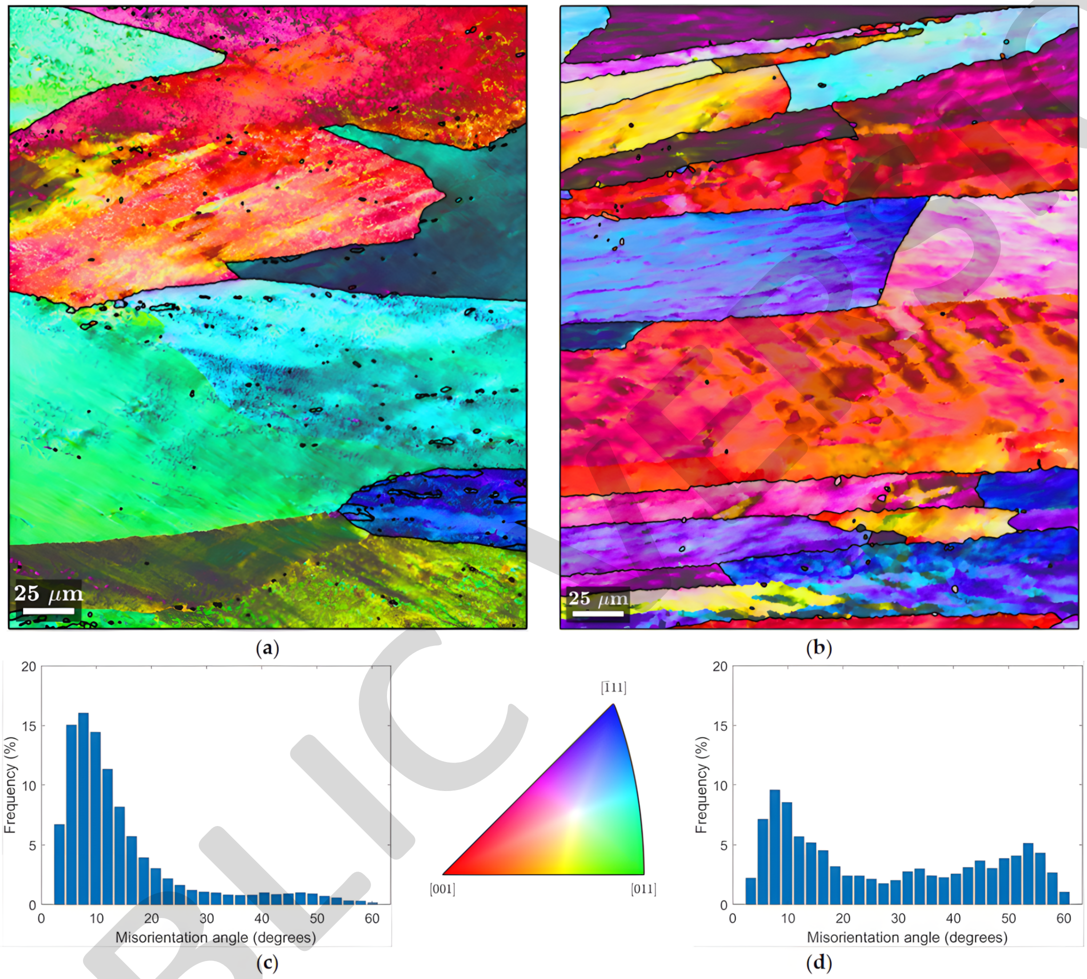

Figure 1.8: EBSD comparison of hot-rolled bulk AA6063 and hot-rolled AA6063 chips, including IPF maps and misorientation distributions [18].

The literature also shows that mechanical performance is not governed by a single parameter. Improved bonding and reduced porosity increase load transfer across former chip interfaces, while deformation substructure, grain refinement and precipitation hardening influence strength.

At the same time, residual oxide networks or unbonded regions can reduce ductility even when strength values appear acceptable [15].

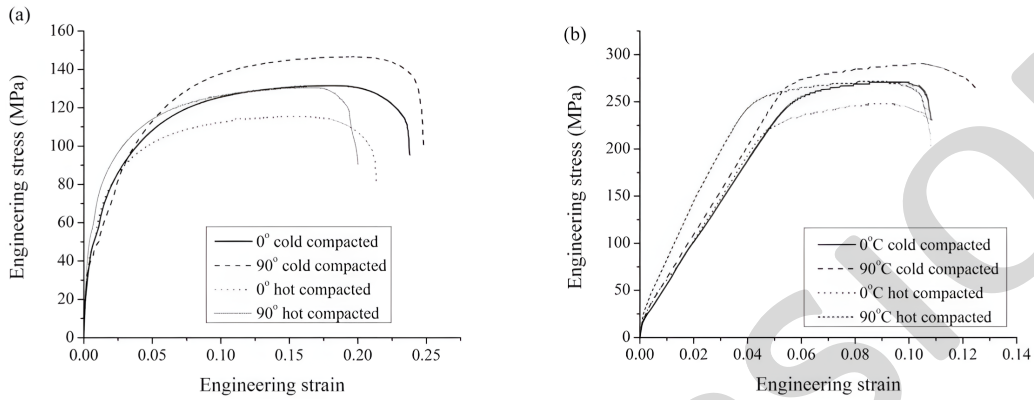

Figure 1.9: Engineering stress-strain curves of hot-rolled sheets produced from compacted aluminium chips after annealing and after T6 heat treatment [16].

Notice regarding the public version

This thesis was conducted as part of a broader ongoing research project.

To protect unpublished and confidential research, this public version contains only non-sensitive material.

Sections containing restricted research results, their discussion, and the corresponding conclusions have therefore been omitted.

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